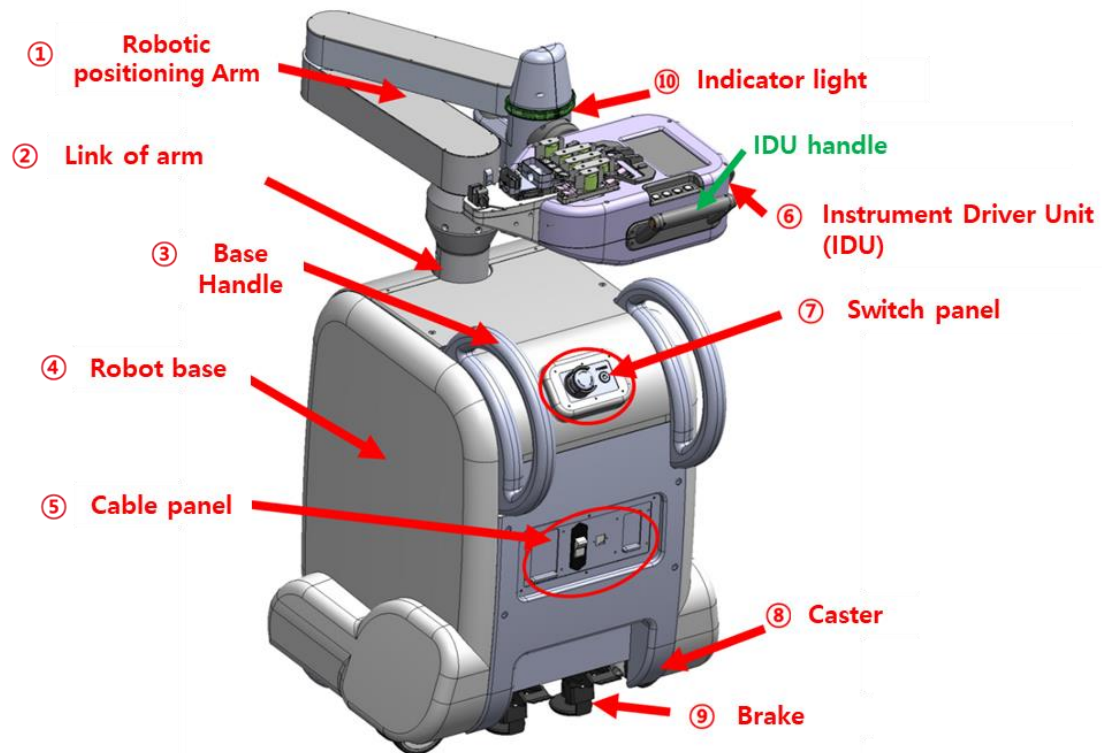
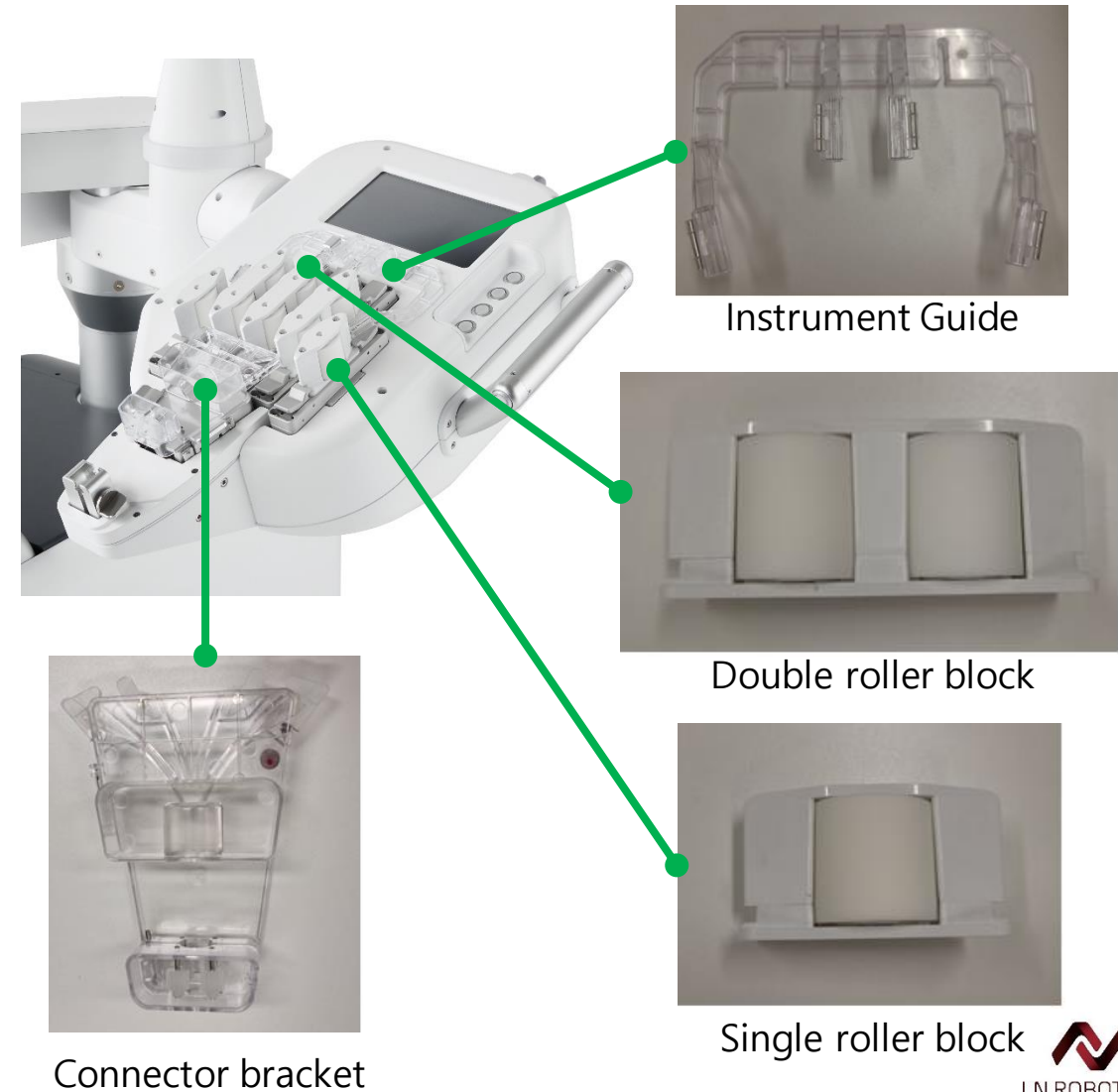


Composition

❖ Procedure robot



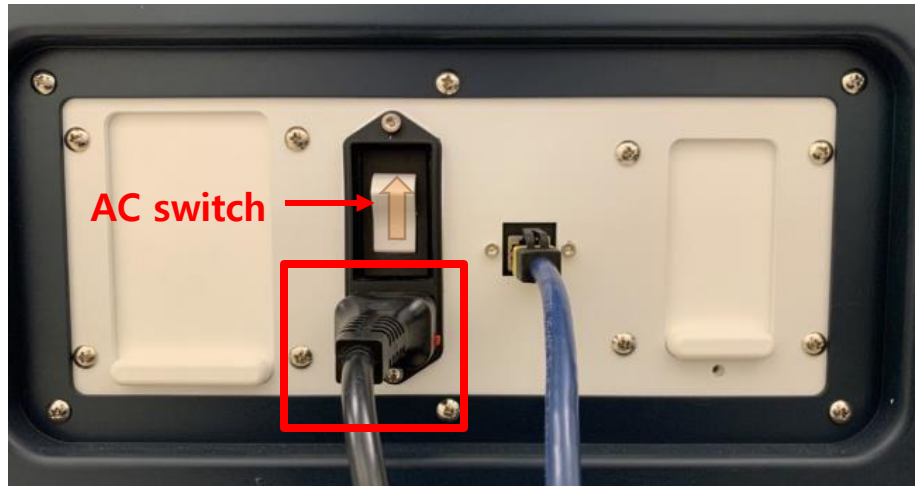
❖ IDU & disposable components



Process for operation

❖ Step 1: Turn on AC Power

- ① Connect power cord and LAN wire to the cable panel of robot base.
- ② Turn on the AC Switch of cable panel.
- ③ When AC power is properly applied, You can smoothly move the robotic positioning arm on the horizontal plane by pressing any button on both sides of the IDU handle.



Process for operation

❖ Step 2: Turn on PC Power

- ① Press the power switches of physician console and procedure robot base.



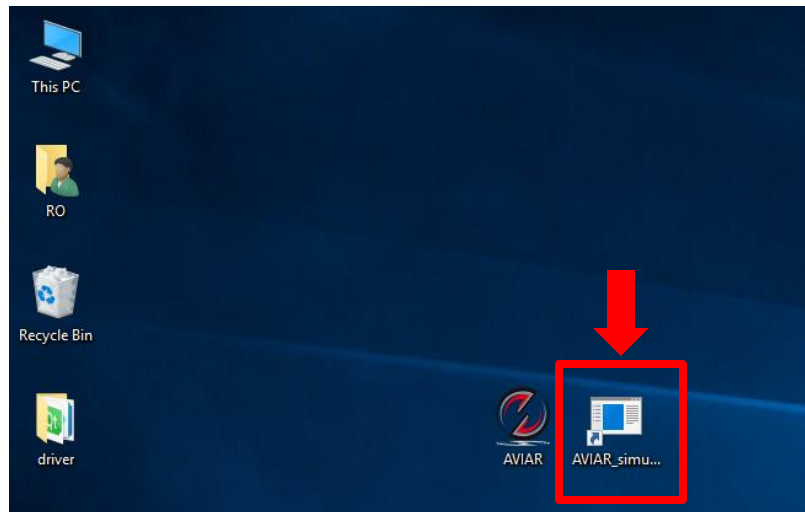
- ② You can use the PC with a monitor by extending the output cable after opening the shutter as shown in the picture below.



Process for operation

❖ Step 3: Run the robot control SW

- ① Double-click the shortcut file, *AVIAR_simulator.exe*, on the desktop.
- ② If there are no issues,
 - each roller block will be moved to the designated position by the motors as soon as run the file (initializing position).
 - you can control switches on the IDU enable to control the vertical motion of robot arm and the angles of IDU.



initializing position



- ① Positioning Arm Vertical Control: Ascending[↕]
- ② Positioning Arm Vertical Control: Descending[↕]
- ③ Instrument Driver Unit Angle Control: Ascending (+)^{↶↷}
- ④ Instrument Driver Unit Angle Control: Descending (-)^{↶↷}



Process for operation

❖ Step 4: operate the IDU using keyboard – for single roller block

① You can select which roller block will move by pressing number keys (you can see which one is selected at the top of the dialog box)

- 1: roller block 1
- 2: roller block 2
- 3: roller block 3
- 4: roller block 4
- 5: roller block 5

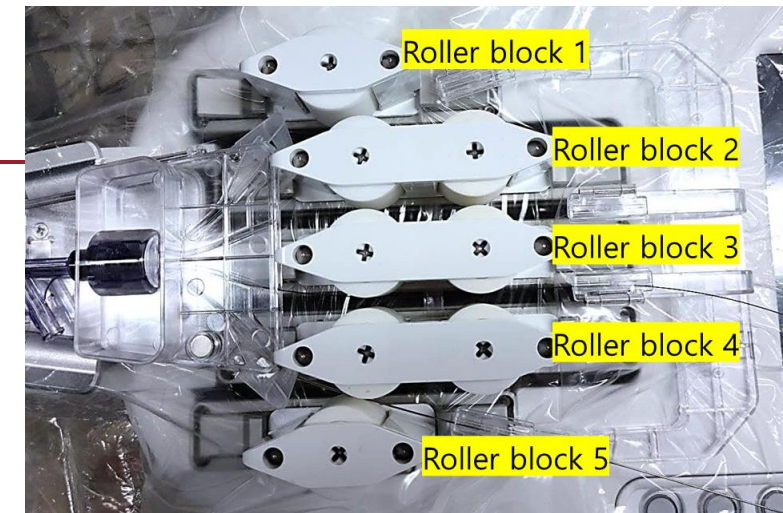
② You can move the selected roller block for

only
roller block 2, 4

- Q: the roller block moves towards the center
- E: the roller block moves towards the edge
- W: the rollers of the roller block rotate clockwise
- S: the rollers of the roller block rotate counter-clockwise

only
roller block 2, 3, 4

- A: the roller block moves downward
- D: the roller block moves upward



Dialog

selected button : _____

1. cassette or Module	2. Keyboard command
1 : Cassette 1	Q : Clamping to center
2 : Cassette 2	E : Clamping to side
3 : Cassette 3	W : Trans. Forwards / Clock (center cassette)
4 : Cassette 4	S : Trans. Backwards / Anticlock (center cassette)
5 : Cassette 5	R : Trans. Clock (side cassette)
6 : Ch 1	F : Trans. Anticlock (side cassette)
7 : Ch 2	A : Rot. Downwards
8 : Ch 3	D : Rot. Upwards
9 : Ch 4	

Reset

Local IP 192.0.0.12 Local port 2020 Start

Remote IP 192.0.0.20 Remote port 54111

Recv Msg Send Msg

Recv msg

Process for operation

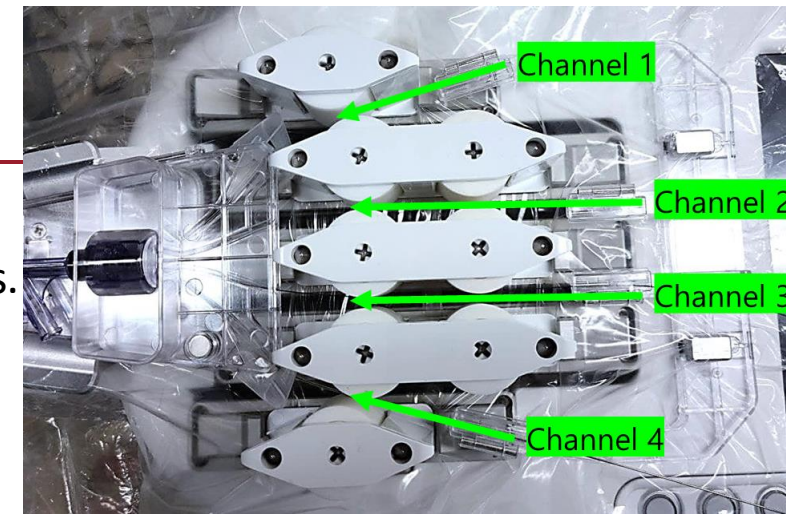
❖ Step 5: operate the IDU using keyboard – for operating with instruments

① You can select which channel of the IDU will use by pressing number keys.

- 6: channel 1
- 7: channel 2
- 8: channel 3
- 9: channel 4

② You can move roller blocks of the selected channel.

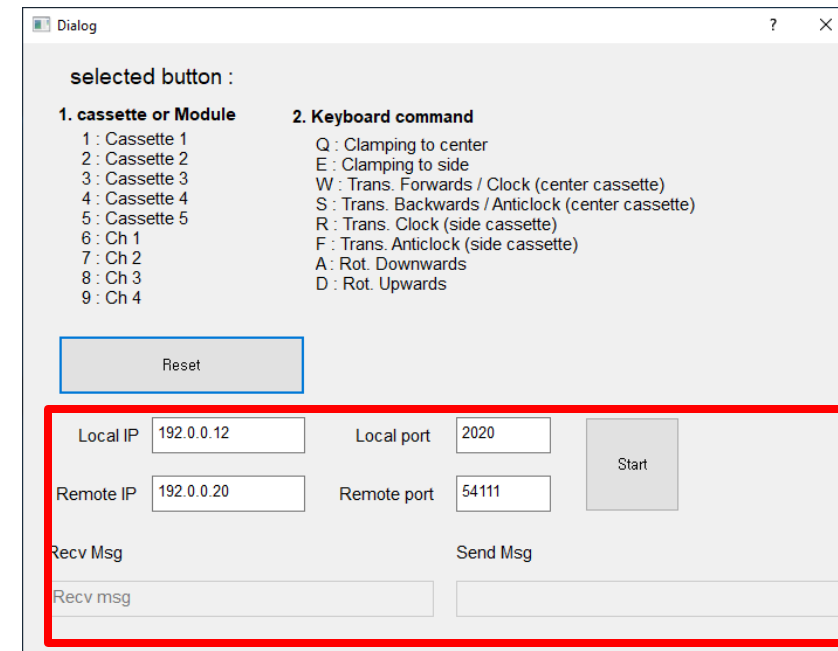
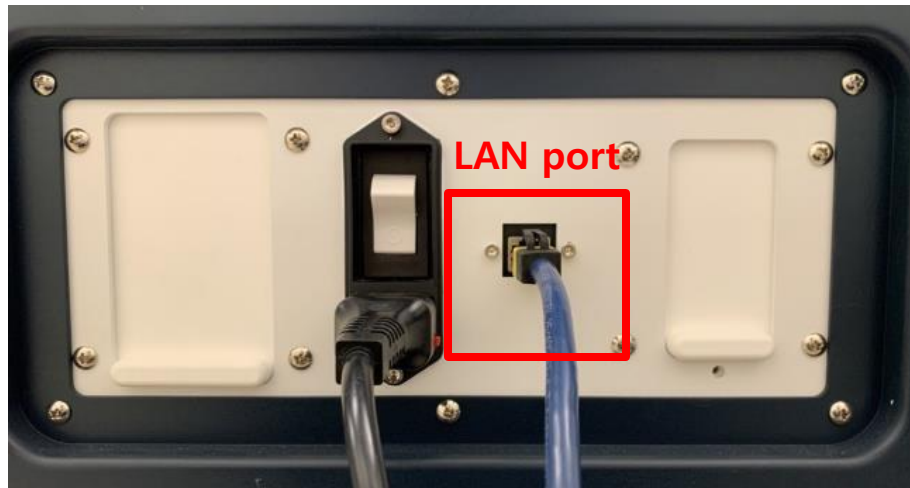
- Q: the roller block moves towards the center to clamp or release instrument (to move roller block 2 or 4)
- E: the roller block moves towards the center to clamp or release instrument (to move roller block 2 or 4)
- W: you can move the instrument forward in the selected channel (Translation forward)
- S: you can move the instrument backward in the selected channel (Translation backward)
- A: you can rotate the instrument in the selected channel (only channel 2, 3)
- D: you can rotate the instrument in the selected channel (only channel 2, 3)



Process for operation

❖ Step 6: operate with UDP communication

- ① Connect the robot and your PC by a LAN cable.
- ② Set the IP addresses and port numbers on the dialog box.
 - Local IP has been set to “192.0.0.12.” Feel free to change it as you prefer.
 - Please check and update the remote IP and port.
- ③ Click the “Start” button. Then, you can receive messages at approximately 500 Hz on your PC.
 - You can see the messages sent from your PC at the bottom of the dialog box.

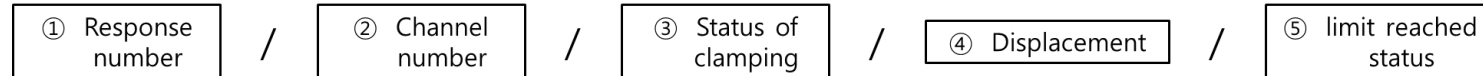


Process for operation

❖ UDP Message protocol

– Robot → external PC

- Communication messages consists of numerical data, each separated by a '/' symbol, encapsulating various information.



① Response number: the message number of the most recently received message

② Selected or operating Channel number

numerical data	0	1	2	3	4
Information	Not selected	Channel 1	Channel 2	Channel 3	Channel 4

③ Status of the selected channel clamped or released

numerical data	0	1	10	100	1000
Information	-	A roller block is moving in direction where the selected channel is closed for clamping guidewire.	Guidewire has been clamped in the selected channel.	A roller block is moving in direction where the selected channel is opened. (towards origin)	Guidewire in the selected channel has not been clamped.

④ Displacement of guidewire (unit: *mm*)

⑤ Status if a roller block reaches the limit height

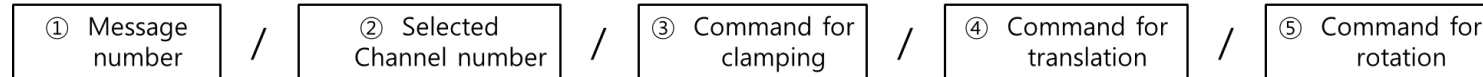
numerical data	0	1
Information	Not reached	Reached

Process for operation

❖ UDP Message protocol

– External PC → Robot

- Communication messages consists of numerical data, each separated by a '/' symbol, encapsulating various information.



① Message number: message number (for use as a timestamp)

② Channel number which was user selected

numerical data	0	1	2	3	4
Information	Not selected	Channel 1	Channel 2	Channel 3	Channel 4

③ Command for clamping

numerical data	0	1	2
Information	-	Release	Clamp

④ Command for translation

- » Speed: 1~15 mm/s
- » Command value = Direction + speed

numerical data	0	Direction		speed
		100	200	
Information	-	Forward	Backward	Integer 1~15

ex.

"112" represents forward movement at a speed of 12mm/s

⑤ Command for rotation

- » Speed: ~360°/s
- » Command value = Direction + speed

numerical data	0	Direction		speed
		100	200	
Information	-	clockwise	Counter-clockwise	Integer ~36 (divided by 10)

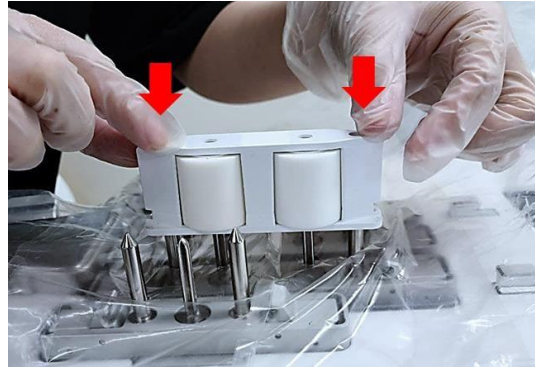
ex.

"109" represents clockwise movement at a speed of 90 °/s

Guide for using instruments

❖ Engaging Disposable Components & Instrument

- ① Engage **roller blocks** at the poles on the IDU and press them until they are fixed in the levers.
 - The secure engagement of the blocks with the levers is important. Without it, there is a risk that the instruments potentially may be impeded due to disengaging the block during rotating the instrument.

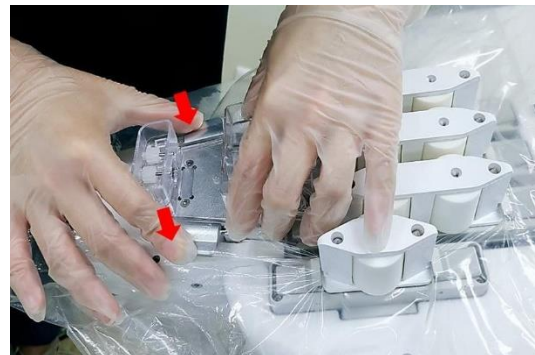


<Mounting a Roller Block>

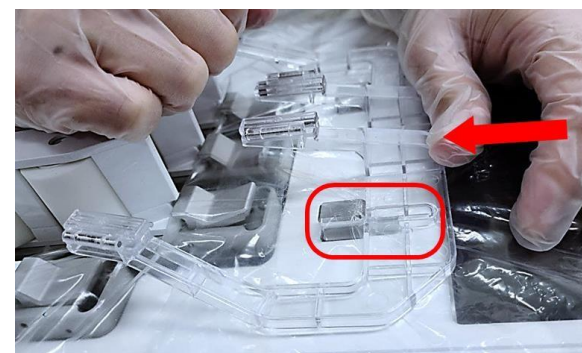


<Mounted Roller Blocks (Total 5 blocks)>

- ② Mount a **connector bracket** by pressing the both levers in front of roller blocks shown the lower left picture.
- ③ Secure an **instrument guide** by pushing it in the direction of the arrow shown in the lower right picture.



<Mounting a connector bracket>

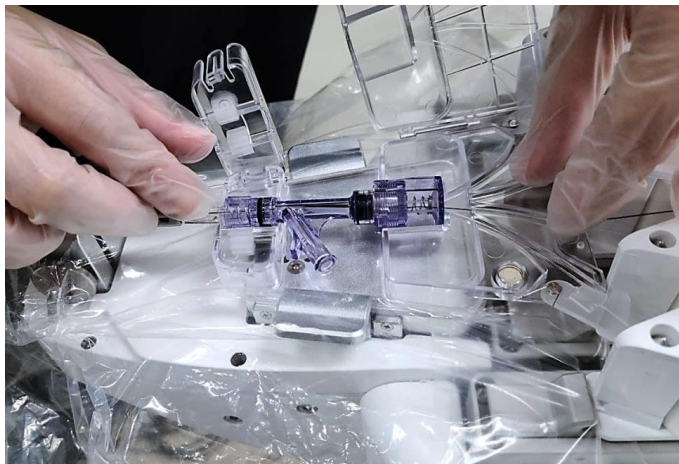


<Mounting an instrument guide>

Guide for using instruments

❖ Mounting Instruments

- ① Open the lid of the mounted connector bracket and mount Y-connector in the connector bracket.
 - Before this step, it would be better if a guidewire is already threaded through the Y-connector.
- ② Accurately position the guide wire, which is threaded through the Y-connector, in the groove of the selected channel on the connector bracket.
- ③ Cover the instruments with the lid of the connector bracket.
- ④ Arrange the guidewire between roller blocks of the selected channel.
- ⑤ Open one of the lids of instrument guide which connected the selected channel.
- ⑥ Position the guidewire on the groove of the instrument guide and cover the lid.



<Mounting Y-connector>



<Guidewire in the connect bracket>

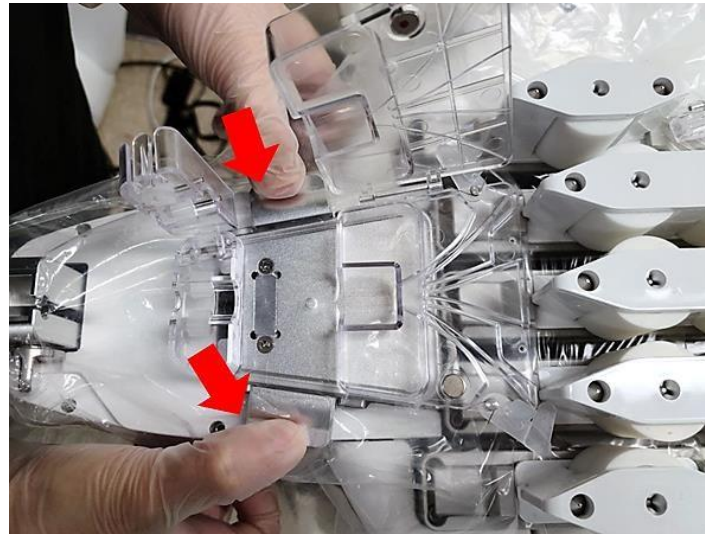


<Guidewire in the connect bracket>

Guide for using instruments

❖ To remove the disposables (after an experiment or for exchanging the disposables)

- ① Open the instrument guide and remove the Y-connector and the guidewire.
- ② Push the instrument guide in the arrow direction, shown in the lower left picture, to detach it from the IDU.
- ③ Press the levers for connector bracket , shown in the lower central picture, to detach the connector bracket.
- ④ Press down the levers, shown in the lower right picture, to detach the roller blocks.



To Use an input device for the PC of the robot

❖ USB port

- You can use the USB port behind the shutter of the cable panel.

